

Do companies bend earnings to avoid reporting a loss?

If earnings were unmanaged, the cross-firm distribution of net income should be smooth. There is no economic reason for a break at exactly zero. A manager near a small loss has discretion — so does the distribution show a **notch**?

16,018

firm-years tested

3,582

distinct firms

2016–25

fiscal years

SEC

Financial Statement Data Sets

37 quarterly SEC archives, 10-K filings only

XBRL-derived flat files, no authentication. Schema verified against the live 2026q1 archive before writing the loader. 113.2 million fact rows scanned.

Why 10-Ks only: a 10-K carries the comparative balance sheet, so one filing supplies both current-year and prior-year total assets — the lagged denominator, from the filer's own comparative column.

Filter	dropped	remaining
raw 10-K firm-years	—	57,407
fiscal years 2016–25 + missing SIC	691	56,716
financial firms (SIC 6000–6999)	14,590	42,126
utilities (SIC 4900–4949)	1,404	40,722
missing net income / bad lagged assets	1,467	39,255
size floor: lagged assets $\geq$ \$10m	8,011	31,244
one filing per firm-year	4	31,240
window $ \text{ROA} \leq 0.10$	15,222	16,018

METHOD

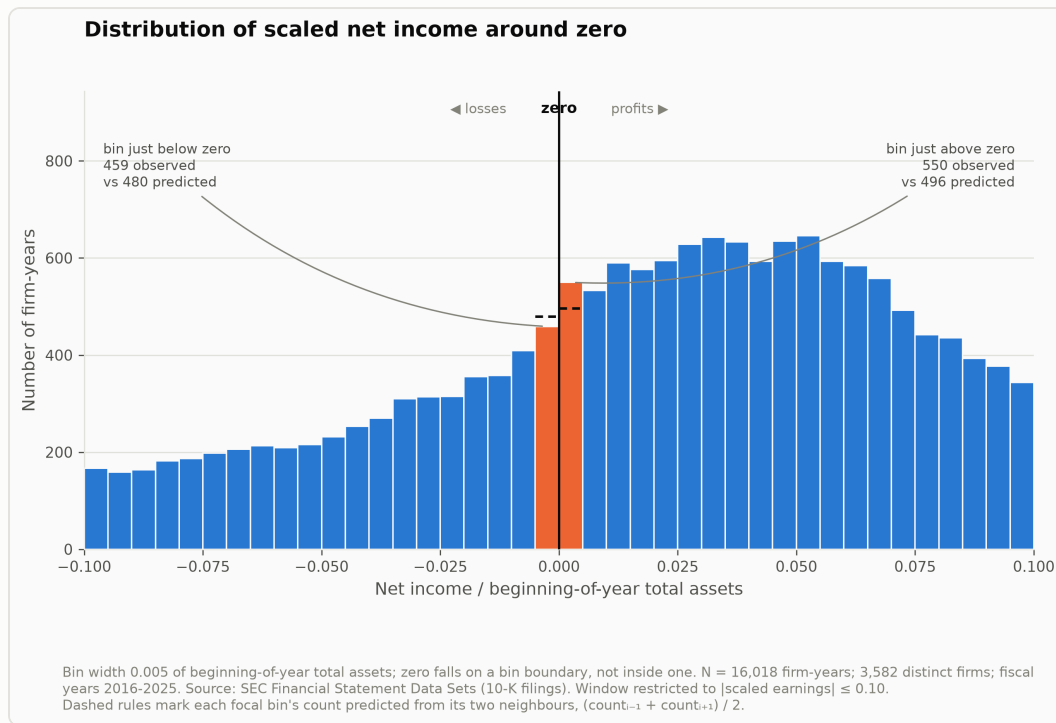
Predict each bin from its neighbours

$\text{expected_i} = (\text{count}_{\{i-1\}} + \text{count}_{\{i+1\}}) / 2$, standardized with the binomial variance. $\text{ROA} = \text{NetIncome}_t / \text{TotalAssets}_{\{t-1\}}$ — beginning-of-year assets, always.

Three commitments, made before looking at any output and enforced in code:

- **Zero on a bin boundary**, never inside a bin — every edge is an integer multiple of the bin width, unit-tested.
- **All three bin widths reported** (0.0025, 0.005, 0.01). A result at one width is not a result.
- **Both sides reported** — deficit below zero *and* surplus above. Finding only one is weaker evidence.

THE HEADLINE



N = 16,018 firm-years, bin width 0.005, zero on a bin boundary. Below zero: 459 vs 480 predicted ($z = -0.79$). Above zero: 550 vs 496 ($z = +1.95$).

ROBUSTNESS 1 — BIN WIDTH

The direction holds. The significance does not.

bin width	z below zero	p	z above zero	p
0.0025	-0.31	0.75	+1.91	0.057
0.005	-0.79	0.43	+1.95	0.051
0.01	-0.87	0.38	+1.73	0.083

Fewer small losses and more small profits than neighbours predict, at every width — the predicted sign. But the deficit below zero is never significant, and the surplus above sits right at the 0.05 boundary.

Size terciles show no concentration in small firms ($z = -0.25 / -0.26 / -0.91$), and the result is flat across size floors from \$0 to \$500m.

ROBUSTNESS 2 — THE SCALAR

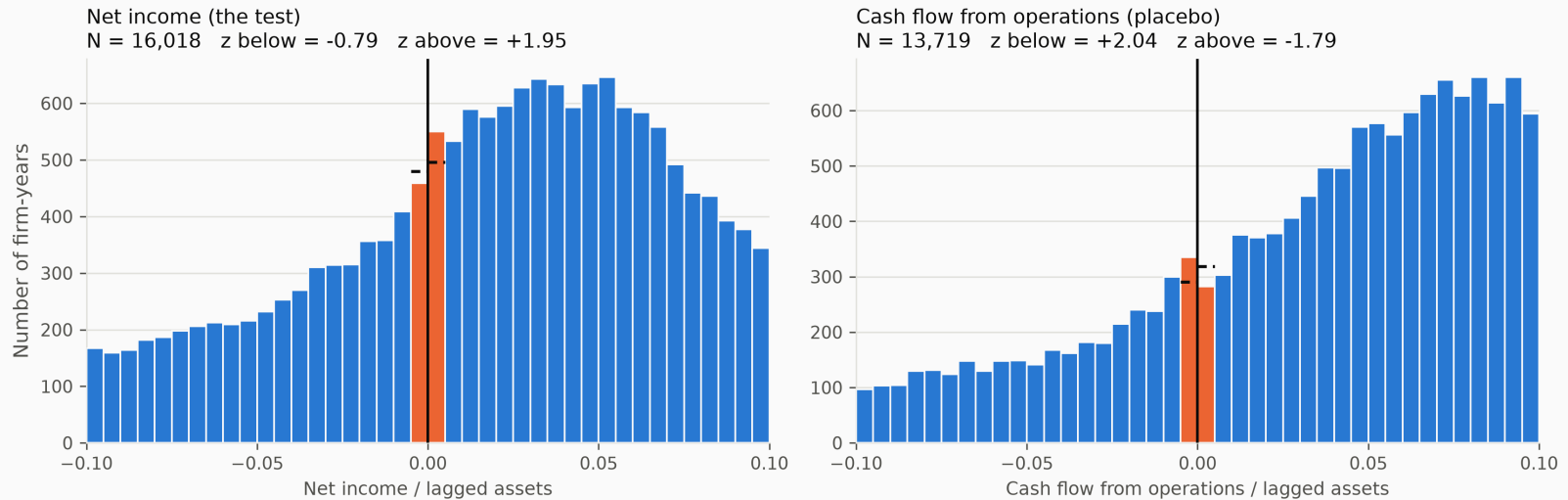
Same firms, three denominators, three answers

denominator	N	z below	p	z above	p
lagged total assets	16,018	−0.79	0.43	+1.95	0.051
revenue	13,452	−2.97	0.003	+4.62	<0.001
lagged book equity	7,177	−1.20	0.23	+1.61	0.11

Revenue scaling produces a clean, significant notch at *all three* bin widths. Assets and book equity produce nothing. **The result is a property of the scaling choice as much as of the earnings** — which is exactly what Durtschi & Easton (2005) argued, arriving here empirically.

ROBUSTNESS 3 — THE PLACEBO

Cash-flow placebo: is the notch specific to accrual earnings?



Bin width 0.005. Both measures scaled by the same beginning-of-year total assets. Source: SEC Financial Statement Data Sets (10-K filings).

Cash flow from operations is much harder to push across zero through accrual timing — so it should show no notch. Instead it shows a **larger break in the opposite direction**: $z = +3.55$ below zero, -3.59 above (bin width 0.01, $p < 0.001$).

BOTTOM LINE

No clear discontinuity — and a useful reason why

- **I did not find significant evidence** of a deficit of small losses in recent SEC filing data. The direction matches the classical result at every bin width; the magnitude does not reach significance.
- **The one significant version is denominator-specific.** Revenue scaling gives a clean notch; assets and book equity do not.
- **The placebo undercuts the null itself.** Neighbour-averaging finds a significant "discontinuity" in cash flow, which cannot have been managed into shape — and it grows with bin width, the signature of density curvature, not a knife-edge break.
- Where a notch appears, **"consistent with earnings management"** is as far as the evidence reaches. This is a distributional anomaly, not a causal finding.
- **Caveats:** pooled firm-years (5.8 per firm) overstate precision; period is 2016–2025 only, so a faded post-SOX notch is compatible with the original result rather than a refutation of it.